PRIYANSHU SHARMA CU24O25O98O (SECTION:B)

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
```

```
import pandas as pd

df = pd.read_csv("/content/titanic.csv")

print(df.head())
print(df.info())
print(df.describe())
```

```
0         892        0       3
1         893        1       3
2         894        0       2
3         895        0       3
4         896        1       3

                                           Name     Sex   Age  SibSp  Parch  \
0                              Kelly, Mr. James    male  34.5      0      0
1              Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                     Myles, Mr. Thomas Francis    male  62.0      0      0
3                              Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1

    Ticket     Fare Cabin Embarked
0   330911   7.8292   NaN        Q
1   363272   7.0000   NaN        S
2   240276   9.6875   NaN        Q
3   315154   8.6625   NaN        S
4  3101298  12.2875   NaN        S
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Survived     418 non-null    int64
 2   Pclass       418 non-null    int64
 3   Name         418 non-null    object
 4   Sex          418 non-null    object
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64
 7   Parch        418 non-null    int64
 8   Ticket       418 non-null    object
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object
 11  Embarked     418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
None
       PassengerId    Survived      Pclass         Age       SibSp  \
count   418.000000  418.000000  418.000000  332.000000  418.000000
mean   1100.500000    0.363636    2.265550   30.272590    0.447368
std     120.810458    0.481622    0.841838   14.181209    0.896760
min     892.000000    0.000000    1.000000    0.170000    0.000000
25%     996.250000    0.000000    1.000000   21.000000    0.000000
50%    1100.500000    0.000000    3.000000   27.000000    0.000000
75%    1204.750000    1.000000    3.000000   39.000000    1.000000
max    1309.000000    1.000000    3.000000   76.000000    8.000000

            Parch        Fare
count  418.000000  417.000000
mean     0.392344   35.627188
std      0.981429   55.907576
min      0.000000    0.000000
25%      0.000000    7.895800
50%      0.000000   14.454200
75%      0.000000   31.500000
max      9.000000  512.329200
```

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
```

```
Age           86
SibSp          0
Parch          0
Ticket         0
Fare           1
Cabin        327
Embarked       0
dtype: int64
```

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

```
/tmp/ipykernel_1297/1933487976.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Age'].fillna(df['Age'].median(), inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
/tmp/ipykernel_1297/3744086084.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
df.drop('Cabin', axis=1, inplace=True)
```

```
print("Before:", df.shape)

df.drop_duplicates(inplace=True)

print("After:", df.shape)
```

```
Before: (418, 11)
After: (418, 11)
```

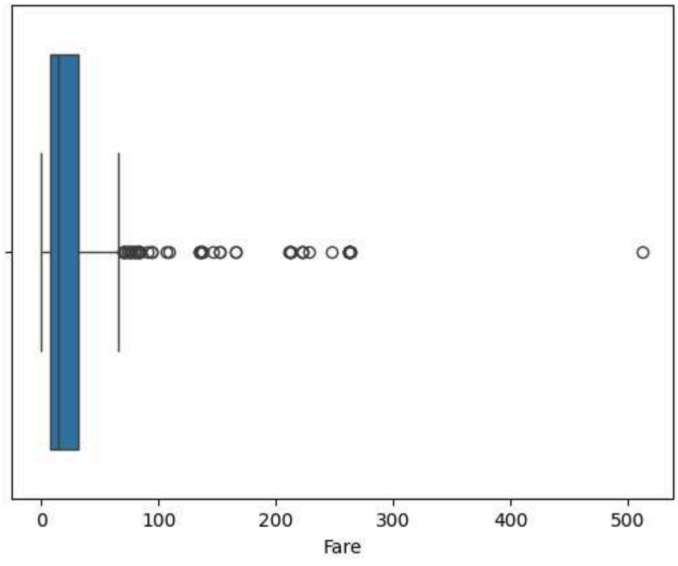
```
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df['Sex'] = encoder.fit_transform(df['Sex'])
df['Embarked'] = encoder.fit_transform(df['Embarked'])
```

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.boxplot(x=df['Fare'])
plt.show()
```




```
Q1 = df['Fare'].quantile(0.25)
```

```
Q3 = df['Fare'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR

upper = Q3 + 1.5 * IQR
```

```
df = df[(df['Fare'] >= lower) &
        (df['Fare'] <= upper)]
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[['Age', 'Fare']] = scaler.fit_transform(
    df[['Age', 'Fare']]
)
```

```
df['FamilySize'] = (
    df['SibSp']
    + df['Parch']
    + 1
)
```

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[-5,-1,0,1,5],
    labels=['Child','Teen','Adult','Senior']
)
```

```
df.to_csv(
    "Titanic_Cleaned.csv",
    index=False
)

print("Dataset Saved Successfully")
```

```
Dataset Saved Successfully
```

Start coding or generate with AI.